

Gauge Absorption: Why Input-Affine FiLM Conditioning of a Mamba Selective Scan Decays to Identity

A Small-Scale Mechanistic Null on a Single RTX 4080

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Abstract

Everything in this paper runs on a single consumer RTX 4080, and the constraint earns its keep: it makes an *exhaustive control apparatus* affordable—an exact bit-identical paired control, forced-active router-supervised escalations, a weight-decay ablation, a toy reference-scan, and a char-Mamba language-model control—and that apparatus, not the hardware, is the contribution. We use it to pin down one training pathology. FiLM-style regime conditioning of a Mamba selective state-space model—a per-block channel-wise affine on each block’s input, meant to steer the input-dependent scan parameters $\Delta, \mathbf{B}, \mathbf{C}$ —is a **decisive null**: under joint training the modulation collapses to identity and cannot be forced active even by direct supervision, on the *weakly-decodable* regime axes a financial world model faces. We explain it: an input-side affine is a *gauge* direction that folds exactly into the block’s own $W_\Delta / W_B / W_C$ projections, so joint training slides it back to identity, with a dataset- and architecture-invariant signature ($\tau \approx 7000$, $R^2 > 0.999$ across FI-2010, Kaggle crypto, and Polymarket) implicating the optimizer, not the data. An output-side placement makes the null *overdetermined*; the toy reference-scan demonstrates the in-kernel boundary; the weight-decay ablation shows the decay is weight-decay-driven over a flat loss; and a language-model control bounds the null by *engaging* the modulation on a strongly-decodable regime. The financial datasets are substrate: no backbone dominates a matched ablation, and the one surviving economic edge is architecture-independent (a logistic reproduces it).

1 Introduction

MOSS celebrates rigorous science at small scale, and this paper is in that spirit: every experiment runs on a **single consumer RTX 4080**, and the ≤ 1 -GPU constraint is what makes the study clean, not what limits it. Selective state-space models (Mamba) (Gu & Dao, 2024) have reshaped long-context sequence modeling, but *which forms of conditioning survive their training dynamics* is poorly understood. We pick one concrete, widely-attempted form—FiLM (Perez et al., 2018) regime conditioning of a Mamba backbone—and diagnose, at small scale, why it fails, then turn the failure into a reusable methods package.

The object of study. We attach a *RegimeFiLM* modulator (a router inferring a soft regime latent, a hypernetwork emitting per-block channel-wise (γ, β)) to a Mamba world-model backbone. Because the selective-scan parameters $\Delta_t, \mathbf{B}_t, \mathbf{C}_t$ are input-dependent linear projections of each block’s input (§2), applying the affine to the block *input* is the natural, kernel-friendly way to push a regime signal *into* the dynamics without touching the fused CUDA kernel—and it is exactly zero-initialized, so at step 0 the modulated backbone is *bit-identical* to the unmodulated one.

The null, and the apparatus that establishes it. Despite that exact control and the strongest activation pressure we can apply—forced-active initialization *plus* direct router supervision on the most defensible regime axis—the modulation decays to identity on every objective and dataset (§3). The mechanism makes the null worth a paper: an input-side affine is a

42 *gauge* direction folding losslessly into the block’s own projections, so the optimizer slides it
 43 back to identity, and the *output-side* placement the gauge argument names does not escape it,
 44 so the null is *overdetermined*. The apparatus that establishes this—listed in the contributions
 45 below—is affordable only because each run is small.

46 **Contributions.** (1) A **decisive, forced-active null** for input-affine FiLM conditioning of
 47 a Mamba selective scan, with an exact zero-init control, on three datasets and a money
 48 objective (§3). (2) The **gauge-absorption mechanism** and its **dataset-invariant** signature
 49 ($\tau \approx 7000$, $R^2 > 0.999$), plus an **output-side placement showing the null is overdetermined**
 50 (App. B). (3) A **small-scale methods package** (§4): the exact-control + escalation protocol,
 51 the toy reference-scan in-kernel demo, the weight-decay ablation, and the char-Mamba LM
 52 control bounding the null to weakly-decodable regimes.

53 2 Setup

54 A selective SSM maps x_t to y_t via a latent $h_t = \bar{\mathbf{A}}h_{t-1} + \bar{\mathbf{B}}x_t$, $y_t = \mathbf{C}h_t$; Mamba makes the
 55 scan parameters input-dependent,

$$\Delta_t = \text{softmax}(W_\Delta x_t), \quad \mathbf{B}_t = W_B x_t, \quad \mathbf{C}_t = W_C x_t, \quad (1)$$

56 so *any* affine transformation of x_t propagates into all three scan parameters. The back-
 57 bone is an $L=4$ stack of Mamba-3 SISO (Lahoti et al., 2026) blocks; a router emits $\alpha_t =$
 58 $\text{softmax}(W_r x_t^{(0)}) \in \mathbb{R}^R$ (R the regime count: $R=4$ on FI-2010/Kaggle, $R=8$ on Polymarket)
 59 and a zero-initialized hypernetwork emits bounded $\gamma^{(\ell)} = 1 + \tanh(\tilde{\gamma})$, $\beta^{(\ell)} = \tanh(\tilde{\beta})$, so
 60 $\gamma=1, \beta=0$ at step 0. The **input-side** site gates the block input $\gamma^{(\ell)} \odot \text{RMSNorm}(x_t^{(\ell-1)}) + \beta^{(\ell)}$
 61 (routing the regime into $\Delta/\mathbf{B}/\mathbf{C}$ via Eq. (1)); a **post-scan** control gates the block output,
 62 sharing the identical modulator; a Switch-Transformer load-balance term (Fedus et al.,
 63 2022) (disabled in the escalations) regularizes the marginal router entropy. Two diagnos-
 64 tics gate every claim: $\text{film}_g = \text{mean}|\gamma - 1|$ (zero iff identity) and the router entropy reg_H
 65 (uniform maximum $\log R$)—load-bearing because a gap measured with an *inert* FiLM is a
 66 distributional artifact, not a conditioning effect.

67 3 The Null, and Why It Holds

68 **Joint training drives FiLM inert.** Across five objectives on two datasets and a money
 69 objective, FiLM collapses to identity: film_g decays toward zero and reg_H sits at its uniform
 70 maximum (Table 1). Where a generalization gap reads positive (direction +0.008, settlement
 71 +0.18), FiLM is provably inert and the treatment is *absolutely worse* in both regimes—the
 72 gap is a distributional artifact whose sign flips with head calibration, which is why the
 73 diagnostics are load-bearing.

74 **Forced-active escalation (single dataset).** The decisive small-scale move is to fight the
 75 optimizer as hard as we can. We force FiLM active ($\text{film}_g = 0.150$ at step 0) and supervise the
 76 router *directly* on the spot efficiency-ratio axis with the strongest settings (LR multiplier 50,
 77 supervision weight 30, decoupled router/FiLM, fed observation volatility). The modulation
 78 still decays monotonically ($\text{film}_g : 0.150 \rightarrow 0.043$, still falling), $\text{reg}_H = \log 8$ throughout. The
 79 capacity is present but unused: a *frozen* router probes to 0.92 agreement on the *volatility*
 80 bucket (the best-decoded axis), yet even there the modulation collapses, while on the
 81 *predictability* axis this escalation supervises the regime decodes only weakly ($\approx 0.25\text{--}0.36$).

82 **Gauge absorption: why identity is a free direction.** The input-side modulation $\tilde{x} =$
 83 $\gamma \odot \hat{x} + \beta$ is applied to a block input the next operations re-project (Eq. (1)). A constant
 84 channel-wise scale folds *exactly* into those projections (for $W_\bullet \in \{W_\Delta, W_B, W_C\}$) as a right-
 85 multiplied diagonal, $W_\bullet(\gamma \odot \hat{x}) = (W_\bullet \text{diag}(\gamma))\hat{x}$, and the shift folds into their biases, with
 86 *no change* to the function the block computes. Identity is therefore a *gauge* direction—a flat,
 87 lossless valley the optimizer slides along—which predicts a decay rate that is a property

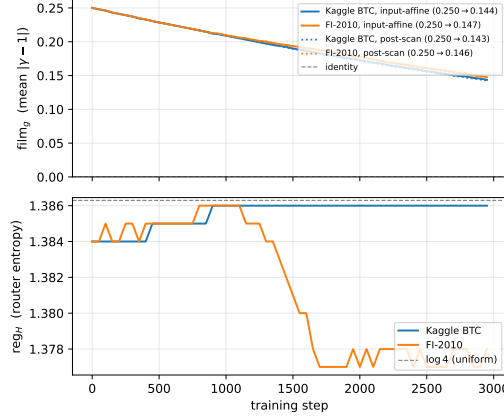


Figure 1: **Strongest-null test, and its placement-invariance.** Forcing the modulation active ($\text{film}_g = 0.250$ at step 0) and supervising the router directly on the efficiency-ratio bucket still decays film_g monotonically toward identity on both datasets (top, solid = input-side), with reg_H pinned at $\log 4$ (bottom). The fit gives $R^2 > 0.999$ with a dataset-invariant $\tau \approx 7000$ and a negative fitted asymptote. The post-scan (output-side) gate (top, dotted) decays along the same trajectory—its scale folds into the bias-free output projection just as the input-side scale folds into $W_\Delta/W_B/W_C$ —so the null is invariant to where the affine is applied (App. B).

88 of the parameterization, not the data. (The fold is exact for the *constant* part of γ ; the
 89 input-dependent part survives only on a decodable regime (§4), so on the weakly-decodable
 90 LOB axes—where the supervised router collapses to uniform— γ reduces to its constant part
 91 and weight decay folds it to identity.)

92 **The strongest-null escalation: a dataset-invariant τ .** We confirm that prediction with a
 93 *cross-dataset strongest-null escalation* (distinct from the single-dataset run above: a stronger
 94 $\text{film}_g = 0.250$ initialization, run on two datasets with the load-balance term disabled). The
 95 modulation decays monotonically on *both* ($0.250 \rightarrow 0.144$ Kaggle, $0.250 \rightarrow 0.147$ FI-2010),
 96 $\text{reg}_H \approx \log 4$ throughout. An exponential fit $a e^{-t/\tau} + c$ matches both at $R^2 > 0.999$ with
 97 near-identical parameters ($a \approx 0.30$, $\tau \approx 7000$ steps); the decay is monotone to the last
 98 logged step (endpoint ≈ 0.15 , still falling) and the fitted asymptote lies at or below identity
 99 ($c \lesssim -0.05$, only more negative when the lower bound on c is relaxed), so it extrapolates
 100 toward identity rather than to a positive plateau. That the rate is essentially *identical* across
 101 two very different markets points to the optimizer, not the data (Figure 1).

102 **The null is overdetermined.** The gauge argument names an obvious next test: gating
 103 *after* the scan. Under the identical escalation, the output-side gate does *not* escape—each
 104 block ends in a *bias-free* output projection, so the output scale folds into it ($\gamma \odot (W_{\text{out}}h) =$
 105 $(\text{diag}(\gamma) W_{\text{out}})h$; $0.250 \rightarrow 0.147$ FI-2010, $0.250 \rightarrow 0.143$ Kaggle, same τ , $R^2 > 0.999$), while the
 106 one non-absorbable component (the additive shift) decays in lockstep under reconstruction
 107 dominance. The null is thus *overdetermined*—by two routes sharing a single driver, weight
 108 decay over a loss that rewards neither: gauge absorption removes every direction the host
 109 weights can represent, reconstruction dominance removes the one they cannot, and a surviv-
 110 ing modulation would have to act *inside* the fused kernel—outside our hardware envelope,
 111 so we delineate the boundary as a location rather than a parameterization (derivations and
 112 the output-side fit in App. B).

113 4 Small-Scale Methods That Establish the Boundary

114 The single-GPU budget is what lets us *lose, repeatedly*, until the mechanism is pinned
 115 down. Three elementary probes (full numbers in App. C) turn the asserted boundary into

a demonstrated one. **(i) Folding (toy reference scan).** An input-affine modulation folds into primed host projections to floating-point precision and decays to identity, while the *same* modulation applied to the projected $\Delta/\mathbf{B}/\mathbf{C}$ *inside* the scan is non-absorbable ($\sim 10^4 \times$ larger fold residual) and **persists** (seeds 0–3)—it survives the very training that decays the affine form. **(ii) Gauge vs. regularization.** With weight decay *off* the input-affine is no longer pulled to identity (its decay time constant blows up $\sim 46\times$), so the decay is **weight-decay-driven over a flat loss**, not earned. **(iii) A language-model control bounds the null.** A 2-layer char-Mamba LM on prose vs. Python, with the domain-supervised regime and the same input-affine FiLM, does *not* collapse: with a strongly-decodable regime (router ~ 0.95 accuracy) film_g rises $0.11 \rightarrow 0.52$. Gauge absorption always removes the *constant* component; the input-dependent component survives only when the regime is decodable enough to be worth encoding—so the collapse is specifically the fate of *weakly-decodable*, drop-in conditioning. (Single-seed toy control; the router $\Rightarrow$ persistence outcome is clear, the plateau value is not load-bearing.)

5 The Substrate Is Not the Subject

The financial datasets are a demonstration substrate; the backbone is a Mamba-3 world model in the DreamerV3/Drama lineage (Hafner et al., 2023; Wang et al., 2024), none of which is the claim—two checks confirm the backbone carries no weight. **(i) No backbone dominates:** a matched-parameter ablation (Mamba-1/2/3 SISO and a Mamba-3-matched pre-norm Transformer) shows no backbone leads on both FI-2010 and Kaggle (single seed, but the cross-dataset gaps are within run-to-run noise; Table 3, App. A)—exactly what a result about how selective scans *train* predicts. **(ii) The surviving economic edge is architecture-independent:** a selective-participation gate turns a directional settlement probability into a survivable high-predictability edge (95% CI $[\text{+\$2,445}, \text{+\$7,747}]$ over 127 markets), but a logistic on the same spot-conditioned features *reproduces and exceeds* it ($\text{+\$6,914}$ vs. $\text{+\$5,147}$)—so the operative variable is the features and the gate, not the sequence model (App. D).

6 Related Work

Scale redundancy and scale-invariance. That a channel-wise affine before a linear map is redundant with that map is a known theme in the normalization literature: van Laarhoven (2017), Hoffer et al. (2018), and Li & Arora (2020) show that weights feeding a normalization layer are scale-invariant, recasting L_2 /weight decay as an effective-learning-rate schedule rather than a capacity control. Our null is an instance in a setting it has not been analyzed: a per-regime, *input-dependent* FiLM folding into the selective scan’s own $W_\Delta/W_B/W_C$. What is new is the input-dependent selective-scan instantiation, the optimizer-implicating dataset-invariant signature ($\tau \approx 7000$, $R^2 > 0.999$), the output-side placement that makes the null *overdetermined*, and that the only kernel-friendly drop-in form is exactly the absorbable one. Mamba-2 recasts the scan as structured masked attention (Dao & Gu, 2024); DeepLOB (Zhang et al., 2019) attains 0.628 macro-F1 on FI-2010 (vs. a 0.261 majority floor), confirming our data carries signal.

7 Discussion

Running everything on one consumer GPU is what makes the apparatus *affordable*, not what makes it valid: the conclusiveness comes from the exact-control-plus-escalation methodology—which would hold on any accelerator—while the single GPU is what lets us run the whole apparatus (the exact zero-init control, the escalations, and the ablations; $\approx 4 \times 10^{16}$ FLOPs total, App. C) cheaply enough to run it all. **Takeaway:** do not gate a selective scan’s input (or output) with a learned affine and expect persistence—it is a gauge the optimizer removes at a data-independent rate; only placements acting on $W_\Delta/W_B/W_C$ *inside* the fused kernel survive, the change that breaks drop-in deployment.

165 **A Diagnostic tables**

Family	Objective	Dataset	Gap (sign, unit)	film_g	Verdict
Volatility/scale	StudentT NLL	FI-2010	≈ 0 nats	$\approx \text{ident.}$	NULL
Order-flow volume	Volume NLL	FI-2010	≈ 0 nats	$\approx \text{ident.}$	NULL
Direction	macro-F1	FI-2010	$+0.008^\dagger$ F1	0.018	NULL
Settlement	BCE log-loss	Polymarket	$+0.18^\dagger$ nats	0.008	NULL
PnL	simulated trade	Polymarket	$-\$627^\dagger$	0.010	NULL

Table 1: RegimeFiLM is inert across every objective. reg_H equals its uniform maximum ($\log R$) in all rows; the Gap column is in each objective’s own units, so only its sign and the inert film_g are comparable. † a non-zero gap whose sign is a distributional artifact, not a FiLM effect. The PnL row’s $-\$627$ is the FiLM A/B gap on the PnL *objective* (treatment–control), *not* an economic total (cf. the $+\$5,147$ settlement total, App. D).

Dataset	Asset / resolution	Gap (pred / vol)	film_g	Verdict
FI-2010	event stream	$\approx 0 / \approx 0$	0.004	NULL
Kaggle BTC	1 min	$\approx 0 / \approx 0$	0.004	NULL
Kaggle ETH	1 min	≈ 0	0.005	NULL
Kaggle ADA	1 min	≈ 0	0.005	NULL
Kaggle BTC	5 min	≈ 0	0.005	NULL
Kaggle BTC	1 s (slice)	≈ 0	0.005	NULL

Table 2: Cross-dataset confirmation of the regime-FiLM null ($\text{reg}_H = \log 4$ in every row): dataset-, asset-, and resolution-independent.

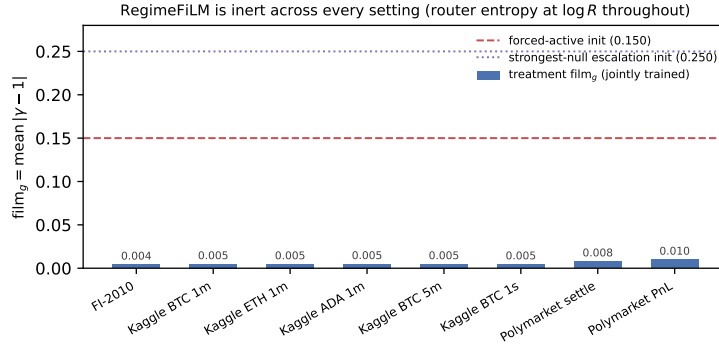


Figure 2: RegimeFiLM is inert across every A/B setting: the jointly-trained film_g (bars) sits near 0.004–0.010 everywhere, while forced-active (0.150) and strongest-null (0.250) initializations are dashed. The modulation never grows from its exact identity initialization.

166 **B Mechanism: the output-side placement in full**

167 The output-side gate applies $y' = \gamma \odot y + \beta$ to the block output. Each Mamba-3 block
 168 ends in a *bias-free* linear output projection $y = W_{\text{out}}h$, so a channel-wise output scale folds
 169 exactly into it, $\gamma \odot (W_{\text{out}}h) = (\text{diag}(\gamma)W_{\text{out}})h$ —the same gauge move as the input side, on
 170 the block’s other face. The additive shift β is genuinely non-absorbable (no output bias,
 171 a nonlinear RMSNorm separating it from the next block’s projections), yet under forced
 172 activation and ER supervision film_g and its additive counterpart $\text{film}_b = \text{mean}|\beta|$ decay
 173 monotonically and in lockstep on both datasets ($\text{film}_g : 0.250 \rightarrow 0.147$ FI-2010, $0.250 \rightarrow 0.143$
 174 Kaggle; film_b tracks within 0.005), $\text{reg}_H \approx \log 4$ throughout. The fit gives the same dataset-
 175 invariant signature ($a \approx 0.30$, $\tau \approx 7065$ FI-2010 / 6598 Kaggle, asymptote $c \lesssim -0.05$ and

Backbone	Params	FI-2010 recon NLL	Kaggle recon NLL
Mamba-1	11.45M	−0.5403	0.0241
Mamba-2	10.20M	−0.5309	−0.0213
Mamba-3 SISO	10.30M	−0.5346	0.0207
Transformer (pre-norm)	9.61M	−0.5041	0.0224

Table 3: Matched-parameter backbone ablation (held-out reconstruction NLL, lower is better). No backbone wins both datasets—Mamba-1 leads on FI-2010, Mamba-2 on Kaggle. Mamba-1 is 11.45M ($\approx 11\%$ over the matched ~ 10.3 M budget, inherited from the parent configuration); excluding it the FI-2010 ordering is unchanged (Mamba-3 SISO -0.5346 still ahead of Mamba-2 -0.5309), and the under-budget Mamba-3-matched pre-norm Transformer is competitive but non-dominant, so the non-dominance is not an under-tuned- or over-budget-baseline artifact.

below identity with the bound relaxed, $R^2 > 0.999$). Two probes localize the failure: the jointly-supervised router is uniform per step ($\approx \log 4$) with near-chance ER agreement (0.25–0.31), and a frozen linear probe decodes the predictability bucket only weakly (0.33–0.36 vs. 0.25) and the volatility bucket somewhat better (0.43–0.47)—yet the volatility-axis gap is also null.

C Methods detail

All toy experiments use the pure-PyTorch reference scan that `mamba_ssm` ships (the slow path the fused-kernel hardware limit does not touch), in seconds on the same single 4080. **Folding:** the recovered projections satisfy $W_{\bullet}^{\text{fit}} \approx W_{\bullet}^{\text{base}} \text{diag}(\gamma)$ to round-off (input-affine residual 1.2×10^{-6}); the in-kernel placement leaves a residual 1.3×10^{-2} because a per-channel scale on the post-softplus Δ cannot be matched by any linear reparameterization of the pre-softplus projection. **Dynamics:** input-affine decays $0.086 \rightarrow 0.010$ (WD on), in-kernel persists $0.093 \rightarrow 0.175$ (seeds 0–3). **Weight-decay ablation:** WD off leaves the input-affine at its forced-active init (toy $0.086 \rightarrow 0.342$; full-model τ from ≈ 6685 , $R^2=0.9999$, to $\approx 3 \times 10^5$, $R^2=0.48$). **LM control:** a 2-layer char-level Mamba LM (built on `mamba_ssm` blocks) on prose vs. Python with a context-aware domain-supervised router and the same input-affine FiLM; router ~ 0.95 accuracy, $\text{film}_g : 0.11 \rightarrow 0.52$. We run Mamba-3 SISO throughout because the MIMO kernel’s dynamic shared-memory footprint exceeds the 4080 cap at every warp-valid (rank, chunk) configuration—the same kernel region a surviving modulation would have to occupy.

Estimated compute (FLOPs). Each pretraining run is ~ 3000 optimizer steps at a 128×64 -token batch ($T \approx 2.5 \times 10^7$ tokens/run); with $N \approx 10.3$ M backbone parameters the dominant cost is $6NT \approx 1.5 \times 10^{15}$ FLOPs/run (~ 1.5 PFLOP, ~ 14 min on the 4080). The full study—the null table, the cross-dataset sweep, the input- and output-side escalations, the weight-decay ablation, and the five-backbone matched ablation (~ 25 runs)—is $\approx 4 \times 10^{16}$ FLOPs; the toy reference-scan and char-LM controls are negligible. Every experiment is thus $\gtrsim 10^3 \times$ below the 10^{20} frontier-scale reference, on one consumer GPU.

Limitations. The *null* is multi-setting and multi-seed; the *constructive* claims that bound it are smaller. The optimizer attribution rests on a τ matched across two datasets plus the within-model weight-decay ablation; the full-model WD-off ablation and the LM control are single-seed (the in-kernel toy is seeds 0–3); the in-kernel “survives” result is at toy reference-scan scale, since the MIMO placement is outside the 4080 envelope; and the substrate is finance-only, so the architecture-universality of the gauge algebra is argued, not demonstrated at language-model scale. None of these bears on the null itself.

D Deflated economic coda

Given the spot path that defines Polymarket settlement, the model’s over-confident-but-directional settlement probability—traded by a selective-participation strategy that bets only when a train-frozen efficiency-ratio gate fires, against a mandatory no-model naive control (which loses $-\$1,008$), summarized by a market-block bootstrap that resamples whole markets—is a survivable, naive-beating edge in the high-predictability regime ($P(\text{profit}) \approx 1.0$, 95% CI $[\$2,445, \$7,747]$ over 127 markets). The edge is architecture-independent: a logistic regression on the same spot-conditioned features reproduces and exceeds it ($+\$6,914$, Brier 0.146, vs. the settlement head’s $+\$5,147$ and a 3-seed EV-head mean of $+\$4,779$; logistic $P \approx 1.0$, exactly 0.9994, over 243 markets). The settlement head’s low-predictability regime is marginal ($+\$51$, $P = 0.63$) and is made robust by a direct expected-value head ($P \rightarrow 0.98\text{--}0.99$, collapsing the high-vs-low PnL spread $6.2\times$ at comparable total PnL). Only the predictability gate is confirmatory; the depth-band and early-market cuts are exploratory, and the reliability/PnL figures are recorded-vintage structural diagnostics. The deterministic conjunction-gated run totals $+\$4,729$ over 1,553 trades (408 markets)—a different denominator from the 645-market settlement total and the 243-market logistic, each labeled where it appears.

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